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B. Tech. Degree III Semester Examination November 2014**SE 304 MANUFACTURING PROCESSES**
(2006 Scheme)

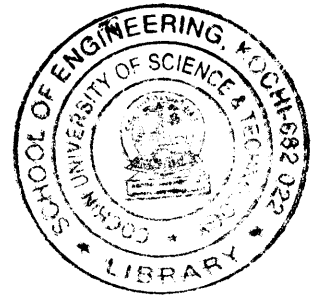
Time: 3 Hours

Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) What is meant by solution hardening?
(b) State the purpose served by quenching and tempering.
(c) Briefly describe the oxy-acetylene welding equipment.
(d) Describe the electron beam welding.
(e) Name the pattern allowances which can be quantitatively specified.
(f) Sketch different types of cores used in founding practices.
(g) List the various processes of metal working.
(h) Why is a built up edge on a tool undesirable?

**PART B**

(4 x 15 = 60)

- II. What are the different tests available for hardness measurement? Compare their individual merits.
- OR**
- III. Explain why tempering follows the quenching process in the heat treatment of steels.
- IV. Give a comparison between TIG and MIG welding process.
- OR**
- V. Explain how the welding quality in submerged arc welding is affected by some of the welding parameters.
- VI. What are the factors which are normally responsible for the production of defects in castings?
- OR**
- VII. How will you compare a cold chamber die-casting process with that of a hot chamber process?
- VIII. Describe the following:
(i) Drop forging
(ii) Press forging
(iii) Machine forging
- OR**
- IX. Explain the following briefly with neat sketches.
(i) Straddle milling
(ii) Dove-tail milling
(iii) Saw milling
